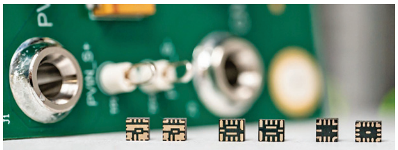
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Industry's smallest 6A power modules

Texas Instruments (TI) has launched power modules that increase power density, efficiency, and reduce electromagnetic interference (EMI). These modules are up to 23% smaller than similar

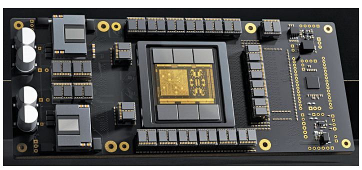


products. This size reduction allows designers in industries such as industrial, enterprise, and communications to reach high performance levels. These modules are the market's smallest 6A power modules, offering a power density of nearly 1A per 1mm^2. This technology enables engineers to achieve high power density and lower temperatures and emissions, reducing board space and system power losses. These enhancements are important in areas like data centres, where power demand is expected to double by the decade's end.

Texas Instruments (TI)
https://www.ti.com

The world's first transformer ASIC

The ASIC dedicated to transformers excels in speed for transformer tasks, outpacing NVIDIA Blackwell GPUs. It



supports transformer models like ChatGPT and Stable Diffusion 3, processing over 500,000 tokens per second. Large AI companies are investing heavily to scale up AI, driven by the benefits of increased computational power. OpenAI and others are preparing for advancements with GPT-5 and beyond, highlighting a future where AI could require resources exceeding the GDP of small nations. Despite GPUs reaching performance plateaus, specialised chips are emerging as essential due to their efficiency and design tailored for transformers, supported by major tech firms.

Etched
https://www.etched.com

First battery-free chip

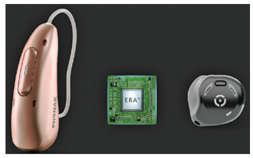
HaiLa Technologies Inc.has partnered with e-peas Semiconductor to develop solutions that combine low-power RF technology and advanced power management integrated

circuits (ICs), aiming to reduce or eliminate the dependence on traditional batteries in various applications. This shift not only cuts operational costs for sensor networks but also reduces environmental impact. The HaiLa BSC2000 RF Evaluation Chip, a breakthrough in monolithic analogue and digital design, utilises HaiLa's passive backscatter technology tailored for Wi-Fi frequencies. By leveraging existing Wi-Fi infrastructure it supports cost-effective deployments in existing settings, further diminishing the need for disposable batteries.

HaiLa Technologies and e-peas Semiconductor
https://www.haila.io

World's first AI-powered hearing aid

Phonak has launched the Audéo Sphere Infinio hearing aid, supported by the Infinio platform. This device is the first to feature a dedicated real-time AI chip, enhancing speech clarity in noisy environments. The hearing aid addresses the challenge of understanding speech against background noise. It uses AI

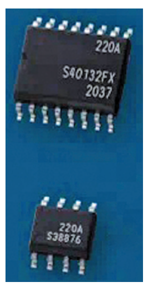


and signal processing technologies to achieve a 10dB improvement in signal-to-noise ratio (SNR), offering improved performance compared to other hearing aids on the market. Key features include sound quality, with a 93% user preference over competitors. It reduces listening effort by 45% compared to previous models and improves speech understanding by 61% in noise when used with the Roger ON accessory. The device offers connectivity, integrating with Bluetooth-enabled devices, and is designed to withstand daily wear and tear.

Phonak
https://www.phonak.com/en-in

Integrated current sensors with reinforced isolation

Melexis has announced a safety qualification (UL/IEC 62368-1) for its MLX91220 (5V) and MLX91221 (3.3V) current sensors, enhancing their application in systems requiring higher voltage isolation. The updated certifications have expanded the use of the sensors in applications like onboard chargers and HVAC/



compressors, while also decreasing the number of components listed on the Bill of Materials (BoM). These sensors now provide isolation to protect users from high-voltage risks, eliminating the need for secondary isolation measures. This